

OEM Manual Excerpt - Centrifugal Pump P-101

OEM Manual Excerpt - Centrifugal Pump Model CP-500

Applicable Asset: P-101, P-102

Section 4: Mechanical Seal Guidance

The standard mechanical seal is rated for continuous duty within the manufacturer's best-efficiency-point (BEP) flow range. Operation away from BEP, including low-flow recirculation, increases the risk of cavitation, which manifests as elevated vibration and accelerated seal face wear. Repeated seal failures on a short interval (see maintenance records for asset P-101) should prompt a review of actual duty cycle and suction conditions against this section before assuming the seal itself is defective.

Section 5: Alignment Tolerance

Shaft misalignment beyond 0.05 mm TIR at the coupling will materially reduce mechanical seal life and is a common root cause of premature seal failure alongside cavitation.

Section 7: Recommended Seal Upgrade Path

For assets experiencing recurring seal failures, the manufacturer recommends the cartridge-style mechanical seal (part family CS-9), which tolerates minor misalignment and transient cavitation better than the standard component seal.